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RESEARCH ARTICLE

Limited plasticity but increased variance in physiological rates across ectotherm populations under climate change

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Abstract

1. Climate change causes warmer and more variable temperatures globally, impacting physiological rates and function in ectothermic animals. Acclimation of physiological rates can help maintain function. However, it is unresolved how variance in physiological rates changes with temperature despite its potential ecological and evolutionary importance.
2. We developed new effect sizes that capture how both the mean and variation in physiological rates change across temperature (based on the temperature coefficient, Q_{10}) and used them to test how acclimation and acute thermal responses vary across aquatic and terrestrial ectotherms using meta-analysis (>1900 effects from 226 species). Comparing both the magnitude of acclimation and changes in variation side-by-side provides unique opportunities for evaluating the importance of plasticity and selection under climate change.
3. We show that variance in physiological rates increases at higher temperatures, but that the magnitude of change depends on habitat. Freshwater and marine ectotherms are capable of acclimation and have the greatest increase in variance. In contrast, terrestrial ectotherms have reduced acclimation abilities and smaller increases in physiological rate. Simulations suggest that these patterns may result from differences in among-individual variation in thermal breadth and optima of performance curves across habitats.
4. Our results highlight the greater vulnerability of terrestrial ectotherms to climate change because of both a lack of acclimation capacity and a limited increase in variance that may provide less raw material for evolutionary adaptation. Considering both acclimation capacity and variance in physiological rates side-by-side is therefore important for understanding how climate change will impact populations.

KEYWORDS
amphibians, ecosystem function, evolutionary physiology, evolutionary potential, fish, invertebrates, reptiles, thermal performance curve

Frank Seebacher and Shinichi Nakagawa contributed equally to this work.

1 | INTRODUCTION

Climate change is expected to result in warmer and more variable thermal environments globally (Suarez-Gutierrez et al., 2023; Ummenhofer & Meehl, 2017). Greater thermal variability is predicted to pose strong selection pressure that leads to genetic adaptation and/or the evolution of adaptive phenotypic plasticity—both of which are considered important for population resilience to human-induced climate change (Chevin et al., 2010; Chevin & Hoffmann, 2017; Chevin & Lande, 2015; Cooke et al., 2021; Seebacher et al., 2015, 2023). Without plasticity or adaptation, high extinction rates are expected unless organisms can migrate to track suitable habitats (Cahill et al., 2012).

Reversible phenotypic plasticity, such as physiological acclimation, is relatively rapid and can be fine-tuned to environmental conditions making it the first 'line-of-defence' against environmental change (Dewitt et al., 1998). For example, physiological rates are known to speed up as temperature increases because of the thermodynamic effects on chemical reaction rates—so called 'acute' temperature responses. However, longer-lasting (days-weeks) temperature increases that move environmental conditions away from thermal optima can be mitigated by acclimation, which adjust reaction rates or the thermal optima itself (Havird et al., 2020; Seebacher et al., 2015). Physiological acclimation is driven by endocrine and epigenetic processes that change the underlying physiology to allow organisms to maintain physiological performance despite changes in the environment (Little et al., 2013; Seebacher & Simmonds, 2019; Taff & Vitousek, 2016). Acclimation therefore alters acute thermal sensitivity to offset the potentially negative effects of acute temperature changes (e.g. higher energetic demands). Acclimation, however, does not necessarily result in complete compensation in response to environmental change (sensu Huey et al., 1999). Rather, increased physiological rates are often only partially compensated such that ectotherms acclimated to, and measured at, warmer temperatures have higher physiological rates than those acclimated to, and measured at, cooler temperatures (Havird et al., 2020; Huey et al., 1999).

Acclimation is expected to evolve in populations experiencing high but predictable environmental variability, and when the fitness costs of plasticity are low (Chevin & Hoffmann, 2017; Dewitt et al., 1998; Reed et al., 2010). Rohr et al. (2018) show relationships between acclimation capacity, latitude and body size suggesting climate could be an important driver of acclimation responses. In addition, distinct patterns of dispersal, habitat use, and costs of plasticity may result in life-history stages diverging in their capacity for acclimation (Rossi et al., 2019). Species occupying terrestrial habitats exhibit weaker acclimation capacities and, therefore, may be particularly vulnerable to climate change given their greater probability of experiencing thermal extremes that overwhelm physiological homeostasis (Gunderson & Stillman, 2015; Hoffmann et al., 2013; Morley et al., 2019; Seebacher et al., 2015). In contrast, marine and freshwater organisms appear to have greater physiological acclimation capacity (Pottier et al., 2022; e.g. Seebacher et al., 2015), possibly because of differences in thermal variability in these environments (e.g. Steele et al., 2019) that selects for differences in plasticity. However, the focus of research up to now has been primarily on mean physiological responses, neglecting how variability in physiological processes might also be impacted by higher temperatures.

As mean physiological rates increase with temperature, it is likely that intrapopulation variability will also be impacted. Positive mean–variance relationships are common across biology, suggesting that, as physiological rates increase with temperature, so too should variability (i.e. Taylor's Law; Giometto et al., 2015). Differences in the shape of thermal performance curves (thermal breadth, maximal performance and thermal maxima) can reflect among-individual variability at higher temperatures, which can also differ between different levels of biological organisation, environmental conditions, and acclimation responses (Angilletta, 2009; Rezende & Bozinovic, 2019; Schulte et al., 2011; Tattersall et al., 2012). Presumably, increases in variation in physiological rates reflect environment-mediated changes to underlying regulatory networks, which can lead to increased variation in phenotypic outcomes (Costanzo et al., 2021; Matthey-Doret et al., 2020). Quantifying levels of among-individual variation in thermal performance curves is important to understand their capacity to evolve, as well as the resilience of populations to environmental change (Careau et al., 2014).

Changes in physiological rate variability are expected to have consequences for the flow of energy within and between populations, communities, and ecosystems (Barneche et al., 2021; Bolnick et al., 2011; Sanderson et al., 2023; Seebacher et al., 2023). Generally, more variable populations are predicted to be associated with broader niches, have increased growth rates, and decreased vulnerability to environmental change, lowering extinction risk (i.e. "portfolio effects," Schindler et al., 2010) (Bolnick et al., 2011; Forsman, 2014; see also, Forsman, 2015; Hart et al., 2016; Pörtner, 2021; Schindler et al., 2010). In addition, if phenotypic and genetic variation in physiological rates are correlated and linked to fitness, reduced phenotypic variation may limit responses to selection and reduce the capacity of populations to evolve (Hoffmann & Sgrò, 2011; Pelletier & Coulson, 2012). Therefore, maintaining intrapopulation variability in physiological rates in a warmer world may be important for population resilience to climate change.

Here, we use meta-analysis to establish the current state-of-knowledge of the extent to which aquatic and terrestrial ectotherms are capable of physiological plasticity. We then developed new effect sizes to quantify how variance in physiological rates change with temperature to ask the following questions regarding acclimation-induced changes in trait means and variances: (1) Does variance in physiological rates change as temperatures rise? (2) Are temperature effects on means of physiological rates greater than changes in variance across aquatic and terrestrial ectotherms? (3) How do changes in trait mean and variance relate to different life-stages, traits, and habitats? (4) Are changes in mean and variance of physiological rates impacted by past climate history? (5) How are variances in physiological rates expected to change under climate change?

2 | MATERIALS AND METHODS

2.1 | Literature collection

We compiled literature on ectothermic animals that measured physiological rates (e.g. metabolic rates, heart rates, enzyme reaction rates) at two or more temperatures after having been acclimated at these temperatures for at least 1 week. We used data from a previous meta-analysis (Seebacher et al., 2015) and updated Seebacher et al.'s (2015) data by extracting data from suitable studies from our own searches that followed the same search protocol. We extracted data from an extra 65 papers (with a total of 238 effects; a 34.03% increase in the number of published articles). For full details on the search protocol, see the Supporting Information, where we also provide a PRISMA flow diagram of our extraction process (Figure S1).

2.2 | Data compilation

We extracted means, standard deviations and sample sizes for physiological rates measured at the two test temperatures that coincided with acclimation temperatures (Figure 1a). If there were more than two temperatures, we chose only the temperatures that fell within the most likely natural range of temperatures experienced by the species in question to be consistent with Seebacher et al. (2015) (Figure 1). We extracted these data from text, tables or figures of a given paper. Data were extracted from figures using the R package *metaDigitise* (Pick et al., 2019). We also recorded the phylum, class, order, genus and species, and the latitude and longitude from where the experimental animals were sourced. For studies that did not provide latitude and longitude for the population, we searched for similar studies by the same laboratory group to identify where the population was likely to have been sourced. If the experimental animals were derived from the wild, we recorded the nearest latitude and longitude of the field collection site. If the animals were sourced from a commercial supplier, we took the latitude and longitude of the supplier. When it was not possible to find latitude and longitude using these methods, we looked up the distribution of the species in question and took the latitude and longitude of the centroid of the species' distributional range.

2.3 | Q_{10} based effect sizes and sampling variances for means and variances

Following Noble et al. (2022), we calculated a series of temperature-corrected effect sizes that compared mean physiological rates ($lnRR_{Q_{10}}$) as well as the variability in physiological rates ($lnVR_{Q_{10}}$) (Figure 1). These effect sizes are similar to the traditional temperature coefficient (Q_{10}), but with formal analytical approximations of their sampling variances. Sampling variances for effect sizes allowed us to make use of traditional meta-analytic modelling approaches.

2.3.1 | Comparing changes in mean physiological rates

To compare mean physiological rates, we calculated the log Q_{10} response ratio, $lnRR_{Q_{10}}$ (Noble et al., 2022) as follows:

$$lnRR_{Q_{10}} = \ln\left(\frac{R_2}{R_1}\right)\left(\frac{10°C}{T_2 - T_1}\right) \quad (1)$$

where, R_1 and R_2 are mean physiological rates at temperatures T_1 and T_2, respectively. Log transformation of this ratio makes the effect size normally distributed. Equation 1 is essentially a temperature corrected equivalent to the log response ratio (lnRR) (Hedges et al., 1999; Lajeunesse, 2011) when the numerator and denominator are measured at different temperatures. This allows comparisons of the means from two temperature treatments directly regardless of the absolute measurement temperatures. The sampling variance for Equation 1 can be computed as follows (as described in Noble et al., 2022):

$$s_{lnRR_{Q_{10}}} = \left(\frac{SD_2^2}{R_2^2 N_2} + \frac{SD_1^2}{R_1^2 N_1}\right)\left(\frac{10°C}{T_2 - T_1}\right)^2 \quad (2)$$

Here, SD_1^2 and SD_2^2 are the standard deviations, and N_1 and N_2 are the sample sizes of the groups measured at T_1 and T_2, respectively (Figure 1a).

2.3.2 | Comparing variance in physiological rates

Nakagawa et al. (2015) proposed analogous effect size estimates to lnRR that allow for comparisons of changes in variance between two groups, the log variance ratio (lnVR) and the log coefficient of variation (lnCVR). Here, we focus on lnVR but derivations for lnCVR, along with re-analyses with lnCVR, are presented in the Supporting Information. In short, lnVR is a ratio that describes the difference in trait variability between two groups. Like lnRR, lnVR can also easily be extended to its Q_{10} analogue (and associated sampling variance) as follows:

$$lnVR_{Q_{10}} = \ln\left(\frac{SD_2}{SD_1}\right)\left(\frac{10°C}{T_2 - T_1}\right) \quad (3)$$

$$s_{lnVR_{Q_{10}}} = \left(\frac{1}{2(N_2 - 1)} + \frac{1}{2(N_1 - 1)}\right)\left(\frac{10°C}{T_2 - T_1}\right)^2 \quad (4)$$

where parameters are defined above. Equations (3) and (4) describe the change in physiological rate variance (Equation 3) normalised to a 10°C temperature change along with its sampling variance (Equation 4).

2.3.3 | Calculating acute and acclimation $lnRR_{Q_{10}}$ and $lnVR_{Q_{10}}$ estimates

Effect sizes can be calculated from samples of organisms measured acutely at two temperatures or after having been acclimated

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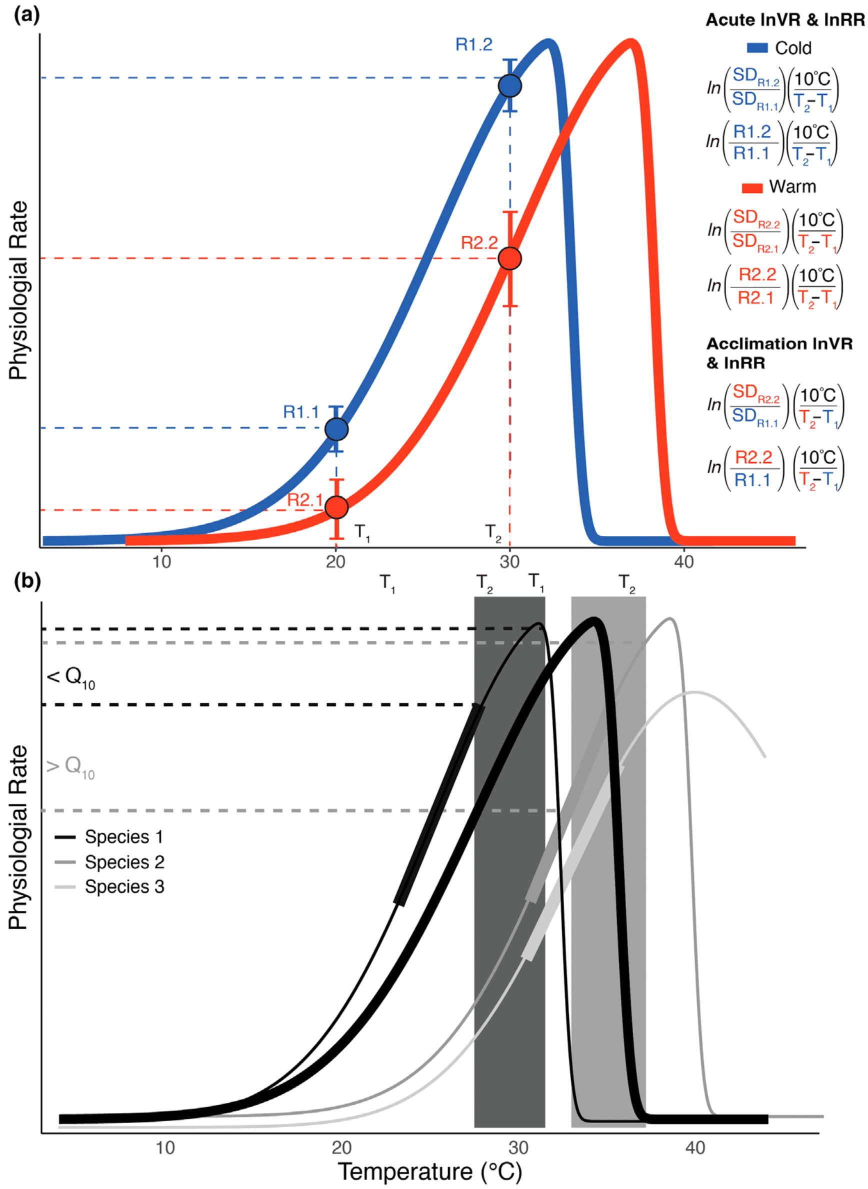

FIGURE 1 Calculations of acute and acclimation $\text{lnVR}_{Q_{10}}$ and $\text{lnRR}_{Q_{10}}$. (a) Two idealised thermal performance curves for animals acclimated at 'cold' ('blue') temperatures and warm ('red') temperatures. Physiological rates are measured for a sample of ectotherms at two different temperatures along the thermal performance curves (T_1=20°C and T_2=30°C) for both curves. At each temperature a mean physiological rate (R) (points) and its standard deviation (SD) (error bars above and below mean) are estimated. R1.1 and R1.2 are the rates and associated SD (subscripted) for the cold acclimated animals at temperature 1 and 2, respectively. R2.1 and R2.2 are the rates and associated SD (subscripted) for the warm acclimated animals at temperature 1 and 2, respectively. An example of how acute and acclimation $lnVR_{Q_{10}}$ and $lnRR_{Q_{10}}$ are calculated from the treatments within the study is provided on the right-hand side of the figure with reference to each of the four possible groups. Two acute effect sizes can be calculated, one for the cold acclimated animals and one for the warm acclimated animals. Acute effects quantify the thermodynamic impacts of temperature on reaction rates whereas acclimated reaction rates measure how much (if at all) these rates are suppressed from having experienced the temperatures chronically. (b) Species are expected, a priori, to vary in their thermal performance curves (thin lines) around an average (thick black line). We restricted our data to areas of each species' performance curve that fell within the natural thermal range of the species (thick lines on each species-level curve). However, given it was not possible to measure the full performance curve for each species some test temperatures within studies may have converged on or moved past the thermal maxima. In such cases, we expected our Q_{10} effect sizes to be smaller as indicated by comparing the black dashed lines to grey dashed lines.

to these same temperatures (Figure 1a). For studies that measure acute and acclimated responses we used the mean, standard deviation, and sample size to derive both acute and acclimation $\text{lnRR}_{Q_{10}}$ and $\text{lnVR}_{Q_{10}}$ estimates. For studies that only measured R_1, R_2, SD_1^2, and SD_2^2 after acclimation we could only calculate acclimation versions of these effect size estimates. Ideally, all studies would have a fully factorial design but this was not always the case making it challenging to compare acute and acclimated responses within studies. Nonetheless, our analytical models are suitable for dealing with missing within-study acute effects (see below). In addition, analysis of a subset of the data where acute and acclimation effects could be compared within studies yields the same conclusion (see Supporting Information). For all effect sizes the higher temperature was in the numerator and the lower of the two temperatures in the denominator. As such, positive effect sizes indicate that the mean (i.e., $\ln RR_{Q_{10}}$) or variance ($\ln VR_{Q_{10}}$) is larger at the higher of the two temperatures (numerator) when standardised to 10°C. When measuring plasticity, it is the difference between $\ln RR_{Q_{10}}$ acute (denoted, $\ln RR_{Q_{10_{acute}}}$) and acclimation (denoted, $\ln RR_{Q_{10_{acclimation}}}$) that captures the degree to which organisms plastically adjust (or acclimate). As done by Seebacher et al. (2015), we consider acute $\ln RR_{Q_{10}}$ and $\ln VR_{Q_{10}}$ as animals measured acutely at both temperatures even though one of the acute measurements is also the acclimation temperature. A better measure of “acute” responses would be to calculate $\ln RR_{Q_{10}}$ and $\ln VR_{Q_{10}}$ on two completely new temperatures but this was not often done in studies. Importantly, our effect sizes, as with Q_{10} more generally, all assume that the effect of temperature on physiological rates (or changes in variance) is log-linear (see Figure 1b and Supporting Information for further discussion). We test and control for any violations of these assumptions in our analysis (see below).

2.4 | Moderator variables

We recorded or derived a series of moderator variables from each study that are expected to have an impact on our effect size estimates. This included the duration of acclimation in days given that acclimation responses may depend on how long chronic temperature exposure occurs. We also recorded if the sample of animals were derived from captive or wild stocks, the life-history stage of the animals used (“adult” or “juvenile”) and the habitat type (“freshwater”, “marine” or “terrestrial”) given that Seebacher et al. (2015) show that these factors can impact Q_{10} estimates. Physiological rate measures varied widely across the studies but could generally be grouped into two broad categories that included whole-organism measures, which all integrate a diversity of physiological and biochemical processes, and biochemical processes (e.g. enzyme reaction rates, proton leak) (Rezende & Bozinovic, 2019; Seebacher et al., 2015). We explore differences across more detailed trait categories in Supporting Information, but note sample sizes are limited for many traits. Traits that could not be categorised into these two we classified as ‘Other’.

2.5 | Meta-analysis

We analysed our data using multilevel meta-analytic (MLMA) and meta-regression (MLMR) models in R (vers. 4.4.2) using *brms* (vers. 2.22.0 Bürkner, 2017, 2018; Stan Development Team, 2021) and *metafor* (vers. 4.6.0 Viechtbauer, 2010). We fit both Bayesian and frequentist approaches to ensure that our results were consistent, and to create orchard plots that more easily convey heterogeneity in effects with prediction intervals (vers. 2.0; Nakagawa et al., 2023; Nakagawa, Lagisz, et al., 2021). Prediction intervals can be interpreted as the range of expected effects from future studies (Noble et al., 2022). In addition, Bayesian methods better protect against type I errors in the presence of complex sources of non-independence (Nakagawa, Senior, et al., 2021; Noble et al., 2017; Song et al., 2020). In all cases, frequentist and Bayesian models resulted in the same conclusions. For our Bayesian models, we ran 4 MCMC chains, each with a warm-up (burn-in) of 1000 followed by 4000 sampling iterations keeping every 5 iterations for a minimum of 3200 samples from the posterior distribution. We used flat Gaussian priors for ‘fixed’ effects (i.e. $N(0, 10)$) and a student t-distribution for ‘random’ effects (i.e. $\text{student}_t(3, 0, 10)$). We checked that all MCMC chains were mixing and had converged (i.e. $R_{hat} = 1$). We also explored the potential for publication bias in our dataset (Nakagawa et al., 2022) but there was no evidence it existed (details in Supporting Information). We report overall meta-analytic means (denoted by μ) and contrasts between meta-analytic means (denoted by β) throughout.

2.5.1 | Multi-level meta-analysis (MLMA) models

We first fit multi-level meta-analysis (MLMA) models (i.e. intercept-only models) for both $\ln RR_{Q_{10}}$ and $\ln VR_{Q_{10}}$, that included study, species, trait type and phylogeny as random effects to account for non-independence and identify sources of variability. We refer to this model structure as “Model 1” in the results. Our MLMA models allowed us to partition the variation in $\ln RR_{Q_{10}}$ and $\ln VR_{Q_{10}}$ among these key sources while accounting for total sampling variance in each. This allowed us to calculate the proportion of total heterogeneity (i.e. I^2_{total}; sensu Nakagawa & Santos, 2012; Noble et al., 2022) along with various I^2 metrics describing the proportion of variance explained by each random effect level (Nakagawa & Santos, 2012). We also present 95% prediction intervals which describe the expected distribution of effects for future studies (Nakagawa, Lagisz, et al., 2021; Noble et al., 2022).

A phylogeny was derived using the Open Tree of Life (OTL) with the *rotl* package in R (vers. 3.1.0) (Michonneau et al., 2016) and plotted using *ggtree* (vers. 3.14.0) (Yu et al., 2017). We resolved all polytomies in the tree randomly using using the *multi2di* function in *ape* (vers. 5.8) (Paradis & Schliep, 2019). Any missing taxa were replaced with closely related species and branch lengths were computed using Grafen's method (using power = 0.7, Grafen, 1989). Models fit using correlation matrices computed with different power (p) parameters (from 0.5 to 1.0) had nearly identical AIC_c. As such, we used an intermediate value of $p = 0.7$. We used the R packages *ape* and *phytools* (vers. 2.3.0) (Revell, 2012) to prune the tree for individual analyses and calculate phylogenetic covariance (or correlation) matrices used in meta-analytic models.

2.5.2 | Multi-level meta-regression (MLMR) models

After quantifying levels of heterogeneity, we fit a series of multi-level meta-regression (MLMR) models to test our key questions. In all models, we included the same random effects as we used in our MLMA models. Acclimation time varied from 4 to 408 days (mean ± SD = 37.98 ± 45.19 days), and terrestrial ectotherms were acclimated for a much shorter duration (mean ± SD = 23.53 ± 15.56 days, n = 125) than freshwater (mean ± SD = 36.81 ± 28.71 days, n = 430) and marine species (mean ± SD = 46.18 ± 67.21 days, n = 313). To control for these differences, acclimation time was mean-centered (mean = 0) and included in all our models, although it was not a strong predictor of effect size variation (Supporting Information, Figure S3).

In addition to the acclimation period, all our models corrected for possible violations of the log-linearity assumption associated with effect size calculations (Figure 1; and see Supporting Information, Figure S2). We predicted that, if $\ln RR_{Q_{10}}$ and $\ln VR_{Q_{10}}$ were not strictly log-linear, there would be a decrease in average effect size for studies applying higher temperature treatments because these temperatures are expected to either converge on or cross the thermal maxima of the performance curve causing reaction rates to decelerate beyond T_{op} (Michaletz & Garen, 2024). Given that our data included a wide range of species and habitats, we also included a random slope of maximum temperature that varied across species because we expected that species would vary in their thermal performance curves, which would be reflected in experimental treatments. We mean-centered the maximum temperature and included it in our models.

Lastly, all models included a random slope of effect type (acute vs. acclimation) to estimate the variance in the magnitude of plastic changes (acute vs. acclimation) across studies. Such an analysis is similar to analyses using an effect size that is a contrast between $\ln RR_{Q_{10_{acute}}}$ and $\ln RR_{Q_{10_{acclimation}}}$ but is more powerful because it allows studies without acute responses to be included (see Supporting Information).

Accounting for these in our meta-regression models, we proceeded to build separate models that tested our core questions. All estimates from our models are therefore conditioned on an average acclimation time (i.e. 37.98 days) and an average maximum temperature (i.e. 23°C) across the dataset. We first tested the extent to which acute and acclimation $\ln RR_{Q_{10}}$ and $\ln VR_{Q_{10}}$ effect sizes varied between habitat types (i.e. terrestrial, freshwater and marine). Models included an interaction between effect type (i.e. acute or acclimation) and habitat (referred to as "Model 2"). Reduced mean $\ln RR_{Q_{10_{acclimation}}}$ relative to $\ln RR_{Q_{10_{acute}}}$ indicates that acclimation to thermal environments results in (partial) compensation of physiological rates (i.e. phenotypic plasticity), whereas no differences between $\ln RR_{Q_{10_{acute}}}$ and $\ln RR_{Q_{10_{acclimation}}}$ indicates that organisms did not acclimate (Havird et al., 2020; Seebacher et al., 2015). In contrast, a difference in $\ln VR_{Q_{10_{acclimation}}}$ relative to $\ln VR_{Q_{10_{acute}}}$ would show that changes in between-individual variation differ between acute responses and acclimation responses.

Second, we tested whether acute and acclimation $\ln RR_{Q_{10}}$ and $\ln VR_{Q_{10}}$ differed between whole-organism versus biochemical traits across habitats by fitting an model with an interaction between type, habitat and trait category (referred to as "Model 3"). A more detailed trait analysis is presented in the Supporting Information. We expected that whole-organism traits would be more likely to maintain variation in physiological function and be less likely to acclimate because whole-organism function relies on a greater number of biochemical reactions each with different thermal sensitivities (Angilletta, 2009; Fields, 2001; Iverson et al., 2020).

Third, we tested whether different life-stages were more or less likely to acclimate by fitting a model for each habitat type and including an interaction between life-stage ('adult' or 'juvenile') and effect type (referred to as "Model 4"). We predicted that acclimation responses would be more likely early in development compared to later in development as this pattern has been shown in previous studies (e.g. Moghadam et al., 2019), but that this should depend on the habitat type given the different constraints faced by different early life stages across major habitat types.

Finally, we used the ERA5 climate model (Hersbach et al., 2020) to test whether the change in $\ln RR_{Q_{10_{acclimation}}}$ and $\ln VR_{Q_{10_{acclimation}}}$ were predicted by climate variability (CV) (see further details in the Supporting Information). We only used $\ln RR_{Q_{10_{acclimation}}}$ and $\ln VR_{Q_{10_{acclimation}}}$ for these models because our predictions were specifically focused on acclimation responses. We fit models that included an interaction between habitat type and thermal coefficient of variability (CV) as moderators (referred to as "Model 5"). We also explored whether environmental predictability explained capacity for acclimation; we estimated predictability as the correlation of temperatures across months at a given location. However, such analyses are challenging to interpret because the temporal scale that is biologically relevant to different organisms will be different making the choice of lag to estimate the correlation difficult to apply across taxa. As such, we report a simple analysis in the Supporting Information but note that it does not differ from our CV analysis.

2.5.3 | Modelling how climate change can impact variance in physiological rates

To explore the potential consequences of the impacts that human-induced climate change may have on variance in physiological rates we fit a model that included a non-linear smoother between latitude and longitude and an interaction between effect type and habitat type while correcting for acclimation time and maximum temperature (referred to as "Model 6"). We used non-linear tensors for latitude and longitude as any response could be complicated by local factors (e.g. altitude). Our model included random effects of species, trait, phylogeny and study. We predicted the expected change in $\ln VR_{Q_{10}}$ for each wild population in our dataset at its specific populations latitude and longitude. We first converted the predicted $\ln VR_{Q_{10}}$ to a 1°C change as opposed to 10°C to better map to relevant changes in temperature coinciding with climate change:

$$\ln VR_{Q_1} = \frac{\ln VR_{Q_{10}}}{10} \tag{5}$$

We then multiplied this predicted change by the change in air and sea surface temperatures at the locations of each population (and species) that are expected under high emissions scenarios in 2080.

2.5.4 | Identifying patterns of among-individual variance in performance curves contributing to variance increases

Changes in $\ln VR_{Q_{10}}$ are expected to depend on differences in the among-individual variation in the thermal performance curves across species (Angilletta, 2009). In other words, we expect performance curves to vary among individuals within a population and this variation is expected to co-vary with habitats (Angilletta, 2009). To understand how differences in thermal performance curve variation correlate with the empirical patterns of variance change we observe with increased temperature, we conducted a simple simulation as a sensitivity analysis to better understand the characteristics of performance curves that could lead to our observed changes in variance across habitats. The simulation varied among-individual variation in performance curves to identify the parameters that could produce the results we observed. To simulate performance curves, we used an asymmetrical Gaussian function (Angilletta, 2009):

$$P_T = 2\epsilon^{-\frac{(T-\delta)^2}{2\sigma^2}} \Phi\left(\alpha \frac{T-\delta}{\sigma}\right) \tag{6}$$

where T is temperature, δ is the optimal temperature (the temperature where performance is maximised), σ the performance breadth, and α the skewness of the curve (see Figure S18 in Supporting Information for example curves). We simulated $n=1000$ individual performance curves by varying the amount of between individual variance on each of the key parameters (δ, σ) in all possible combinations from 0.01 to 2. We also varied α, but this did not impact our conclusions and so we kept among-individual variation fixed for each simulation (at 0.01). From the population of performance curves, we took the standard deviation at two temperatures (18 and 28°C) to calculate $\ln VR_{Q_{10}}$ and identify potential parameter space that could produce observed patterns in our empirical data.

3 | RESULTS

3.1 | Data summary

The final dataset included a total of 91 freshwater (fishes = 48, molluscs = 4, amphibians = 19, reptiles = 8, arthropods = 10, and a single crustacean and nematode species), 90 marine (fishes = 47, annelids = 2, molluscs = 21, echinoderms = 7, reptiles = 1, arthropods = 10 and a single crustacean and cnidarian species) and 45 terrestrial species (annelids = 1, molluscs = 5, arthropods = 14, reptiles = 12 and amphibians = 12 along with a single tardigrade species) (Figure 2).

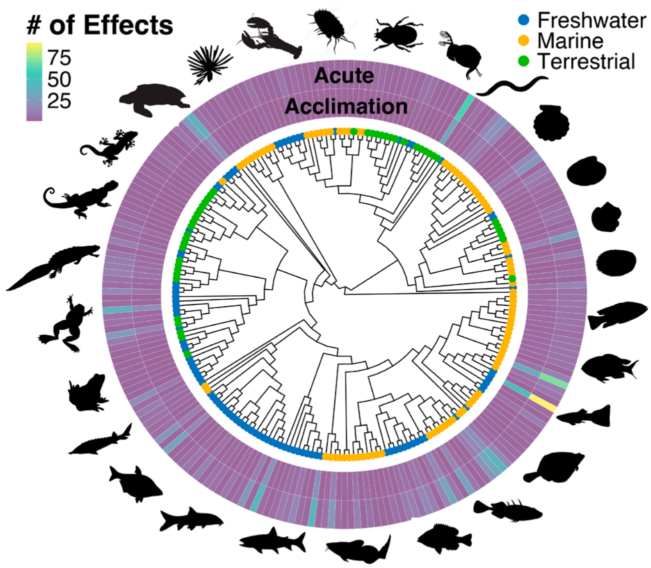

FIGURE 2 Phylogenetic distribution of acute and acclimation $\ln RR_{Q_{10}}$ and $\ln VR_{Q_{10}}$ estimates across major habitats. The total number of acute and acclimation $\ln RR_{Q_{10}}$ and $\ln VR_{Q_{10}}$ effect sizes are indicated by the coloured bars, and the colouring at the tips of the phylogeny indicates marine, freshwater and terrestrial habitats. Silhouettes are only representative taxa of major clades within the tree.

We had more data on acute thermal responses ($n=1115$) compared to acclimation responses ($n=798$) because acute responses were reported for each of the two acclimation temperatures (Figure 2).

Most of the effect size estimates came from measurements of metabolic rates (both resting and maximal—$N_{species}=190$, $N_{effects}=1023$), metabolic enzyme rates ($N_{species}=61$, $N_{effects}=798$) and whole-organism performance traits (i.e. measures of locomotor speed and endurance—$N_{species}=73$, $N_{effects}=321$).

3.2 | Terrestrial and aquatic ectotherms differ in their capacity to acclimate, but acclimation does not depend on life-history stage

Results from "Model 1" (see "Meta-Analysis" above) show that effect heterogeneity was high (only 2.85% of the variance was the result of sampling variability, 95% CI: 2.38% to 3.32%), and most variance was explained by the specific study and type of trait (Study: 29.41%, 95% CI: 20.78% to 38.49%; Trait Type: 29.35%, 95% CI: 19.97% to 39.53%). Evolutionary relationships among taxa and species ecology (i.e. species random effect) explained little variation in acute and acclimation responses (Species: 2.39%, 95% CI: 0.01% to 8.1%; Phylogeny: 2.89%, 95% CI: 0% to 12.94%). These patterns were similar for $\ln VR_{Q_{10}}$ (see Supporting Information, Figure S15).

Physiological rates increased more with temperature in terrestrial ectotherms ($\mu=0.63$, 95% CI: 0.5 to 0.75) compared to marine ($\mu=0.52$, 95% CI: 0.41 to 0.64) and freshwater ectotherms ($\mu=0.56$, 95% CI: 0.45 to 0.65), but did not differ significantly between aquatic and terrestrial habitats (differences between average $\ln RR_{Q_{10}}$;

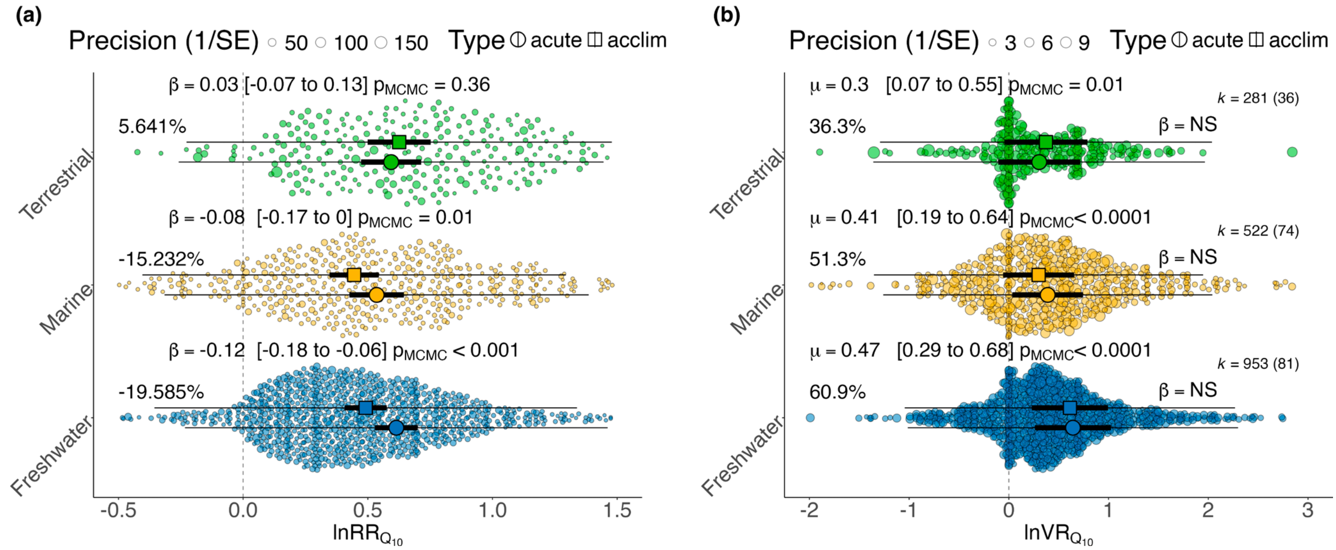

FIGURE 3 Meta-analysis results for different habitats. In both panels, thick bars are 95% confidence intervals (CI) and thin bars 95% prediction intervals (PI). β values are the contrasts between acute and acclimation means within each habitat. μ values are the overall meta-analytic means averaged across acute and acclimation types within each habitat type. In both cases, their 95% CI's are indicated within square brackets and raw effects are weighted by their precision (inverse sampling variance). p_{MCMC} values are the posterior probability of the contrast or overall meta-analytic mean being different from zero. (a) Mean acute and acclimation $\ln RR_{Q_{10}}$ across ectotherms in marine, freshwater, and terrestrial habitats. Overall mean physiological rates (μ) across the habitats are provided in the results for simplicity and only contrasts between acute and acclimation $\ln RR_{Q_{10}}$ are shown. Percentages refer to the percentage change in physiological rates between acclimation and acute $\ln RR_{Q_{10}}$. (b) Mean acute and acclimation $\ln VR_{Q_{10}}$ across ectotherms in marine, freshwater and terrestrial habitats. Percentages refer to the percentage change in physiological rate variance for a 10°C temperature change. For both plots, k = total number of effect size estimates while the numbers in brackets indicate the number of species. Sample sizes are the same for panel (a and b). For ease of visualisation, all the raw data plotted for both acute and acclimation type effect sizes are presented as circles.

Terrestrial - Marine: $\beta = 0.11$, 95% CI: −0.02 to 0.24, $p_{MCMC} = 0.1$; Terrestrial - Freshwater: $\beta = 0.07$, 95% CI: −0.03 to 0.18, $p_{MCMC} = 0.19$) ("Model 2"). However, capacity for acclimation depended on the habitat. Ectotherms in marine and freshwater environments showed partial compensation of physiological rates (Figure 3a) amounting to reduced $\ln RR_{Q_{10_{acclimation}}}$ of 19.59% (95% CI: −28.97 to −10.18) in freshwater and 15.23% (95% CI: −29.26 to 0.21) in marine environments. In contrast, terrestrial ectotherms showed no acclimation with a 5.64% increase in $\ln RR_{Q_{10_{acclimation}}}$ (95% CI: −10.46 to 23.6, Figure 3a).

Acclimation capacity did not vary consistently by life-history stage with no differences in $\ln RR_{Q_{10_{acclimation}}}$ and $\ln RR_{Q_{10_{acute}}}$ between adult and juveniles (overall contrast: -0.04 95% CI: −0.22 to 0.32, $p_{MCMC} = 0.58$). Averaging over acute and acclimation effects there were also no differences between adults and juveniles within habitats (Adult-Juvenile: Terrestrial: -0.07, 95% CI: −0.39 to 0.2, $p_{MCMC} = 0.7$; Marine: 0, 95% CI: −0.21 to 0.22, $p_{MCMC} = 0.97$; Freshwater: 0, 95% CI: −0.12 to 0.12, $p_{MCMC} = 0.95$; "Model 4"; Figure 4a–c).

3.3 | Variation in physiological rates increases but to a greater extent in aquatic compared to terrestrial ectotherms

Variance in physiological rates ($lnVR_{Q_{10}}$) showed an increase with increasing temperature across all habitat types (Figure 3b). Overall, there was a 36.27% (95% CI: 7.51 to 73.59, $p_{MCMC} = 0.01$) increase in physiological rate variance for terrestrial ectotherms, a 51.28% (95% CI:21 to 89.48, $p_{MCMC} = <0.01$) increase in variation for marine ectotherms and a 60.93% (95% CI: 34.05 to 97.92, $p_{MCMC} = <0.0001$) increase in variance for freshwater ectotherms across 10°C (Figure 3; results from "Model 2").

Physiological rate variance increased significantly more in freshwater compared to terrestrial ectotherms for acute responses ($\beta = 0.21$, 95% CI: 0 to 0.41, $p_{MCMC} = 0.05$), but not for acclimation responses because increases in rates were dampened by acclimation resulting in smaller increases in variance ($\beta = 0.11$, 95% CI: −0.12 to 0.34, $p_{MCMC} = 0.35$). While marine ectotherms had larger increases in variance compared to terrestrial ectotherms these were not significant (Acute: $\beta = 0.18$, 95% CI: −0.07 to 0.41, $p_{MCMC} = 0.14$; Acclimation: $\beta = 0.01$, 95% CI: −0.25 to 0.27, $p_{MCMC} = 0.91$) (Figure 3b). Marine and freshwater habitats did not differ in the extent of variance increases at higher temperatures (Acute: $\beta = 0.03$, 95% CI: −0.17 to 0.23, $p_{MCMC} = 0.76$; Acclimation: $\beta = 0.1$, 95% CI: −0.1 to 0.29, $p_{MCMC} = 0.34$). There were no differences between $\ln VR_{Q_{10_{acute}}}$ and $\ln VR_{Q_{10_{acclimation}}}$ within any habitat (Figure 3b).

$\ln VR_{Q_{10}}$ values from our simulations matched our empirical results in particular areas of parameter space (Figure 5). For a given among-individual variance in thermal breadth, terrestrial ectotherms are predicted to have lower among-individual variance in thermal maxima compared to marine and freshwater ectotherms (Figure 5). In contrast, terrestrial ectotherms are expected to have

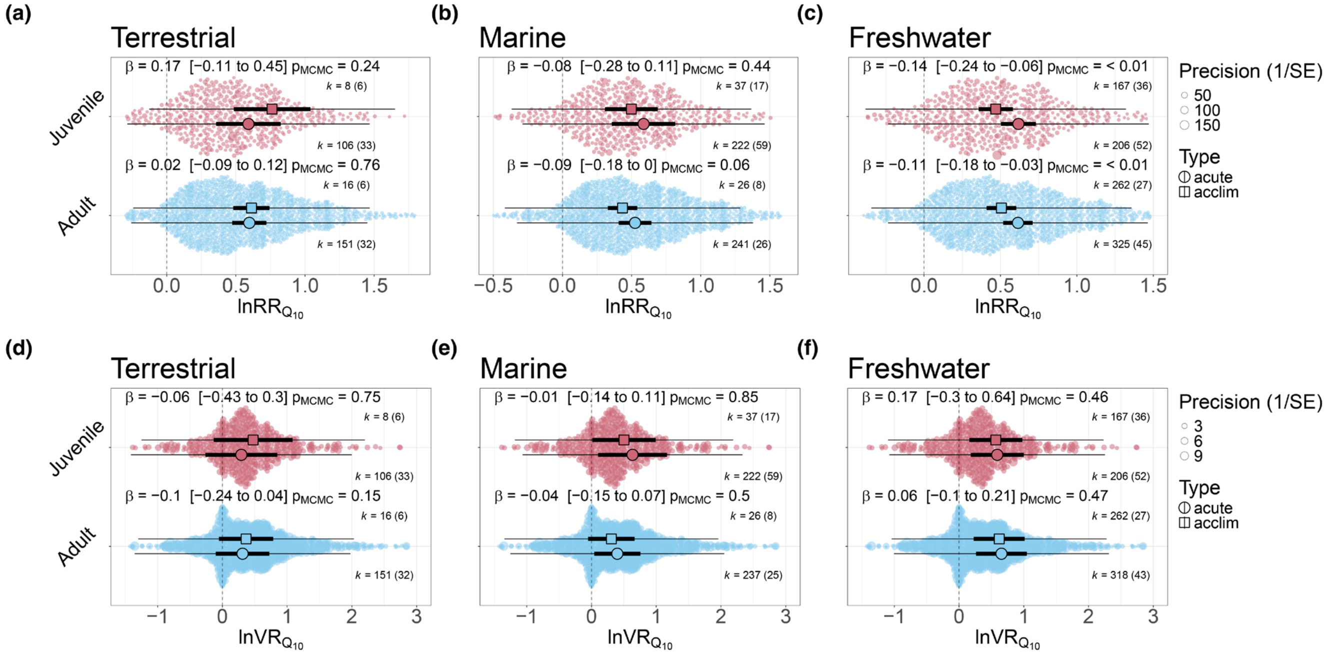

FIGURE 4 Meta-analysis results for different life stages. Estimated mean acclimation and acute $\ln RR_{Q_{10}}$ (a–c) and $\ln VR_{Q_{10}}$ (d–f) for adult and juvenile life-history stages for terrestrial (a and d), marine (b and e) and freshwater (c and f) ectotherms. Across all plots, thick bars indicate 95% confidence intervals and thin bars indicate 95% prediction intervals. Raw effects are weighted by their precision (inverse sampling variance). k = total number of effect size estimates while the numbers in brackets indicate the number of species. For ease of visualisation, raw data for both adult and juvenile life-history stages are presented but points are not distinguished by different symbols. β values are the contrasts between acute and acclimation means within each life stage. p_{MCMC} values are the posterior probability of the contrast being different from zero.

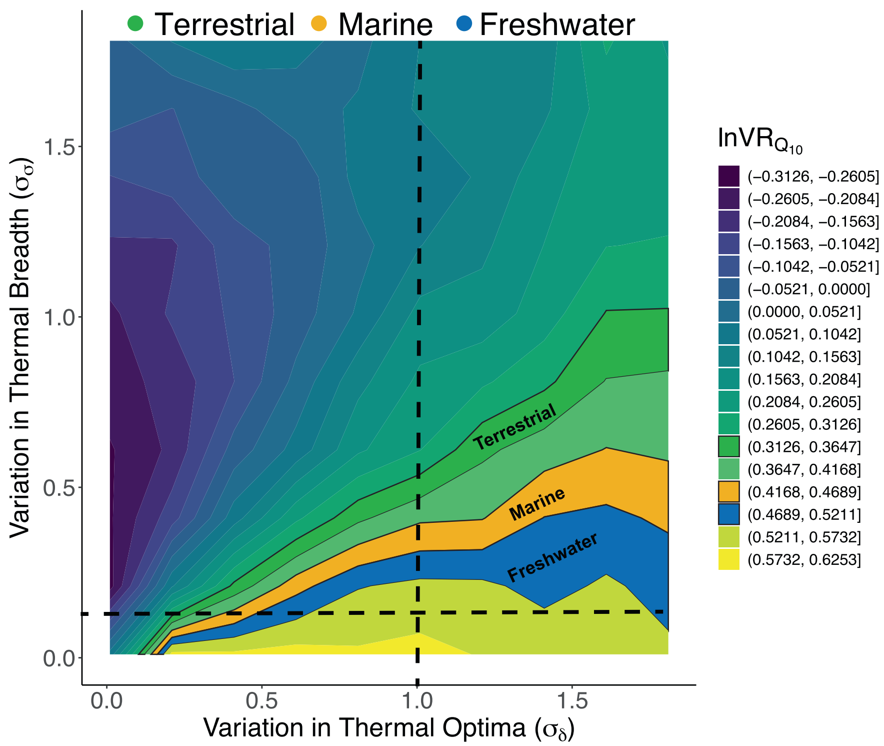

FIGURE 5 Performance curve simulations for the expected $\ln VR_{Q_{10}}$ when varying among-individual variance in thermal breadth (σ_{δ}={0.01, 2}) and thermal maxima (σ_{σ}={0.01, 2}) while fixing the rate variance constant (σ_{α}=0.01). In all simulations, population parameters were δ=35, σ=9, α=−15 and n=1000 individuals were simulated for each combination of σ_{δ} and σ_{σ}. The parameter space that matches the observed mean $\ln VR_{Q_{10}}$ from our meta-analysis for terrestrial (green), marine (orange) and freshwater (blue) ectotherms is labelled and highlighted. Dashed lines indicate the relative differences between the three habitat types when holding one variance parameter constant.

higher levels of among-individual variance in thermal breadth when controlling for among-individual variance in thermal maxima (Figure 5).

Each life-history stage exhibited the same pattern of variance change in each of the habitats (Adult-Juvenile contrasts: Marine: β=0, 95% CI: −0.37 to 0.38, p_{MCMC}=0.98; Freshwater: β=0.03, 95%

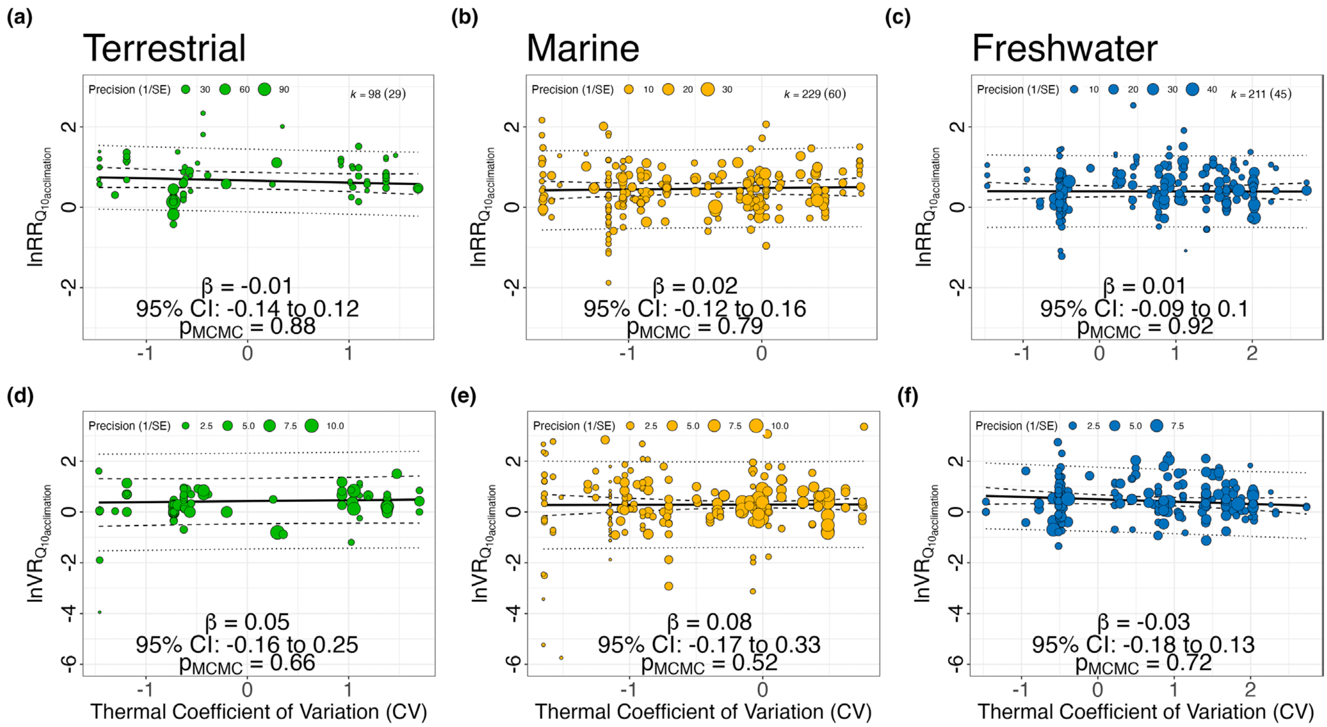

FIGURE 6 Past climate variability did not predict acclimation responses. Predicted mean acclimation (thick black line) $\ln RR_{Q_{10_{acclim}}}$ (a–c) and $\ln VR_{Q_{10_{acclim}}}$ (e–g) as a function of the thermal coefficient of variation (CV) for wild populations across marine, freshwater and terrestrial habitats. Dashed lines indicate 95% confidence intervals and dotted lines indicate 95% prediction intervals. Raw effects are weighted by their precision (inverse sampling variance). Model slope (β) along with the 95% CI and p_{MCMC} values for the slopes are shown for each habitat.

CI: −0.16 to 0.23, p_{MCMC} = 0.72; Terrestrial: β = −0.03, 95% CI: −0.52 to 0.42, p_{MCMC} = 0.93, overall across habitats: β = 0, 95% CI: −0.45 to 0.38, p_{MCMC} = 0.92), with no differences between acute and acclimation effect types ("Model 4"; Figure 4).

3.4 | Past climate does not influence acclimation capacity or expected change in variance

Thermal variability (i.e. *CV*) experienced by a population in the past did not explain acclimation capacity (Figure 6a–c) or changes in physiological rate variance (Figure 6d–f) among terrestrial, marine or freshwater populations ("Model 5").

3.5 | Changes in physiological rate variance under climate change

Measurements of acute and acclimation responses from wild ectotherms were much less common than from captive populations ($N_{species}$ = 134, from 188 wild populations). Globally, there was a clear bias towards species in the Northern Hemisphere (Figure 7a–c). Projected changes in physiological rate variance were highly variable across the globe, however, variance was predicted to increase at all locations. Latitude and longitude explained variation in these responses with models containing smoothers being supported over models with just main effects of latitude and longitude (ΔwAIC = 2.9).

Using the ERA5 climate model, predictions of current global changes in physiological rate variance were generally conservative with our model explaining ~46% of the variation in the observed data (R^2 = 0.43, 95% CI: 0.33 to 0.51). Climate change is predicted to result, on average, in a 28.75% increase in variance for freshwater systems (95% CI: 15.35 to 47.62%, p_{MCMC} = <0.0001), a 15.67% increase in marine systems (95% CI: 0.62% to 30.31%, p_{MCMC} = <0.0001) and a 13.01% increase in terrestrial systems (95% CI: 7.11% to 19.47%, p_{MCMC} = <0.0001) under a RCP8.5 climate scenario (Figure 7d). All results are derived from "Model 6".

4 | DISCUSSION

Understanding acclimation capacity and how variation in physiological rates change across populations and species is important for predicting the ecological and evolutionary consequences of climate change (Bolnick et al., 2011; Bush et al., 2016; Chevin et al., 2010; Chevin & Hoffmann, 2017; Sanderson et al., 2023; Seebacher et al., 2023; Urban et al., 2023). While most of our data are from vertebrates and fish, we show that both acclimation responses ($\ln RR_{Q_{10}}$) and increases in physiological rate variance at warmer temperatures ($\ln VR_{Q_{10}}$) of ectotherms varied across habitats. Our results uncover an hitherto unrecognised dynamic where the benefits of partial

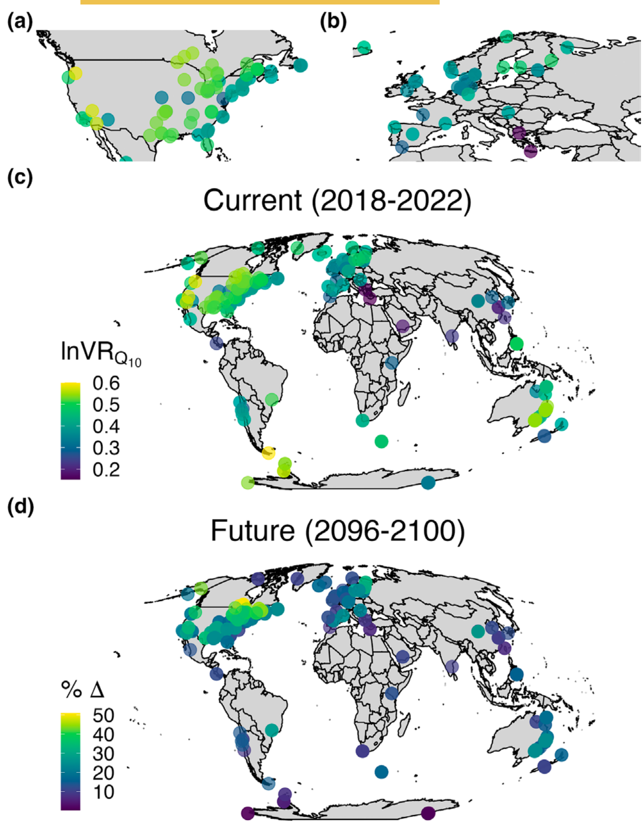

FIGURE 7 Potential effects of climate change on trait variance. Model predictions for the expected change in lnVR$_{Q_{10}}$ across the globe for terrestrial, marine and freshwater ecthotherms. Predicted change in physiological rate variance for each population based on current temperatures (average from 2018 to 2022; a–c) as well as the expected change from current temperatures based on future temperature predictions (average from 2096 to 2100, d). Future climate predictions are the increase in variance expected under a RCP8.5 climate scenario relative to current climate conditions (% change).

acclimation are paralleled by increases in trait variance that depend on habitat in ways that may have impacts on how ectotherm populations will be able to adapt to increased temperatures.

4.1 | Acclimation capacities vary among habitats but are often still limited

We show that the capacity for acclimation of physiological rates differs across habitats. Our findings confirm previous results that quantify the different capacity of terrestrial, marine and freshwater ectotherms to acclimate (Gunderson & Stillman, 2015; Morley et al., 2019; Seebacher et al., 2015). Our analysis also confirms findings by Seebacher et al. (2015), Gunderson and Stillman (2015) and Morley et al. (2019) that all show a general inability of terrestrial ectotherms to physiologically acclimate. These consistent results are interesting given the different physiological traits measured in these meta-analyses (e.g. thermal limits versus physiological rates).

The change in acclimation Q_{10} we found in our expanded dataset was similar to Seebacher et al. (2015) for freshwater organisms (~17%), but higher in marine ectotherms (decrease of 16% versus ~10% in Seebacher et al., 2015), and lower in terrestrial ectotherms (increase of ~6% compared to an ~8% decrease in Seebacher et al., 2015). The difference observed in terrestrial ectotherms between studies may be due to additional data from terrestrial species added in our analysis and to the use of newly derived Q_{10} effect sizes that allowed us to control for sampling variance. Greater capacity for acclimation in aquatic organisms may be the result of fewer opportunities for behavioural thermoregulation in aquatic environments making physiological remodelling important for maintaining homeostasis (Gunderson & Stillman, 2015; Morley et al., 2019). Importantly, even though marine and freshwater ectotherms were capable of partial acclimation, on average, the effect size was small (amounting to Q_{10} dropping from ~1.8 to 1.6), suggesting that acclimation provides limited scope for aquatic ectotherms to adjust their physiology to higher temperatures.

4.2 | Increased variability in physiological rates across habitats: Adaptive potential of physiological processes in the face of climate change?

Contrary to acclimation capacity, variance in physiological rates increased across habitats, with effect sizes being 3–5 times larger than those observed for acclimation of mean trait values. Mechanistically, it is unclear what exactly is contributing to the increased variation in physiological rates at higher temperatures, but it is likely the result of increased among-individual variability in how biochemical, cellular and physiological processes function at higher temperatures to maintain homeostasis (Angilletta, 2009; Fields, 2001; Schulte et al., 2011; Somero, 1995; Tattersall et al., 2012). Higher temperatures increase membrane fluidity, affecting electrochemical gradients and impacting protein structure and function (Fields, 2001; Somero, 1995; Tattersall et al., 2012). Such challenges (among others) may expose among-individual variation within a population. Indeed, there is considerable variation in acclimation capacity among individuals, which would also increase variance in thermal performance curves within populations (Loughland & Seebacher, 2020; Schulte et al., 2011).

Importantly, increased variance in physiological rates was not equal among terrestrial, marine and freshwater ectotherms, with increases in variance being higher in freshwater ectotherms (~60% increase/10°C) compared to terrestrial ectotherms (~36% increase/10°C). One possible hypothesis for the differences in variability we observed across habitats could be that among-individual variation in key parameters affecting the shape of thermal performance curves differ between habitats (Angilletta, 2009; Huey & Kingsolver, 1989; Rezende & Bozinovic, 2019; Tattersall et al., 2012). Our simulations suggest that theoretical and observed lnVR$_{Q_{10}}$ match when thermal performance curves have different among-individual variance in thermal maxima and breadth across habitats making this hypothesis plausible. Such patterns across habitats are expected given that terrestrial ectotherms should be adapted to more extreme and variable thermal environments. Theoretical models also suggest that populations with greater temporal environmental variability exhibit greater thermal breadth (Lynch & Gabriel, 1987). However, we did not find support that thermal variation co-varied with $\ln VR_{Q_{10}}$ (see below), as would be expected. The relevance of analyses of thermal variability will depend on temporal variation in temperature that is biologically relevant—a challenging feat across diverse taxa, but worthy of future investigation.

Our results further highlight the potential vulnerability of terrestrial ectotherms to climate change. Assuming that changes in variation in physiological rates are underpinned by genetic variation, and that there is a genetic correlation with fitness, smaller increases in physiological variance could limit adaptation in terrestrial habitats more than aquatic habitats in the future (Hoffmann & Sgrò, 2011; Urban et al., 2023). For example, under climate change we expect an increase in variance in physiological rates of only ~13% in terrestrial habitats whereas for freshwater habitats we expect variation in physiological rates to increase by ~30%. Importantly, responses to selection will also depend on the magnitude and direction of genetic covariances with other traits, which need consideration. There will obviously be limits to variance increases, and we predict that organisms closer to their upper thermal limits (CT_{max}) will have lower $\ln VR_{Q_{10}}$ values compared to those farther away from CT_{max}. Some evidence points to possible differences across habitats in upper thermal limits already (Gunderson & Stillman, 2015; Pinsky et al., 2019), making this a fruitful future question to explore.

4.3 | Plasticity and variance in physiological rates do not differ between life stages

Acclimation capacities are expected to differ between life-stages because of distinct patterns of dispersal, habitat use and behaviour that force earlier life stages to cope with more variable environmental conditions which can also lead to developmental constraints on how physiological systems respond later in life (Angilletta, 2009; Martin, 2015; Noble et al., 2018; O'Dea et al., 2019; Pottier et al., 2022; Sinclair et al., 2016; Stearns, 1976). In addition, plastic responses are also expected to be costly (Angilletta, 2009; Dewitt et al., 1998); such costs can be magnified in later life, reducing the capacity for plasticity (e.g. Rossi et al., 2019). These processes can also result in changes to intrapopulation variation in physiological rates at higher temperatures, but the direction of change between early and adult life stages is likely to depend on the costs of adjusting physiological processes, energy reserves at different life stages, and the extent to which early life experiences constrain plasticity.

Despite these expectations, our analysis does not show any significant differences between early and late life acclimation capacities and little change in the variance in physiological rates across habitats. This may not be too surprising given that such responses are likely context or trait-dependent (Carter & Sheldon, 2020; Moghadam et al., 2019). The lack of differences we observed may be because both juvenile and adult animals occupy similar thermal niches, disperse to a similar extent and exhibit comparable thermoregulatory behaviours, making physiological responses to temperature similar. A focus on collecting more detailed information on behaviour, dispersal and thermal environments experienced by different life stages is likely to provide a more complete picture on when plasticity differs. We would also encourage more empirical focus on this question and its potential ecological and evolutionary implications.

4.4 | Past climate does not influence capacity for physiological acclimation or changes in variance

Theoretical models predict that plasticity should evolve in populations experiencing greater environmental variability (spatial or temporal), particularly when fluctuations are predictable over time to make environmental cues reliable (Chevin et al., 2010; Chevin & Hoffmann, 2017; Lande, 2009; Murren et al., 2015; Reed et al., 2010). Higher spatial and temporal heterogeneity in terrestrial habitats (Steele et al., 2019) therefore suggests that plasticity is more likely to evolve in terrestrial environments. However, if thermal variability is too high and unpredictable, the rates of acclimation decrease and there are increased costs associated with re-modelling physiological processes (Angilletta, 2009) it would instead be expected that phenotypes are canalised during development (Angilletta, 2009; Leung et al., 2020, 2023; Loughland & Seebacher, 2020; Rescan et al., 2022; Seebacher et al., 2015). The lack of acclimation in terrestrial ectotherms we observed is consistent with the latter hypothesis and is supported by other meta-analyses of heat tolerance (Barley et al., 2021; Gunderson & Stillman, 2015) suggesting that there are costs to being plastic or that the environmental signals are insufficient to trigger endocrine and epigenetic mechanisms that lead to plasticity when environments are not predictable (Leung et al., 2020).

Whether population capacity for acclimation is related to the thermal variability (or predictability) it experiences is equivocal. We show no relationship between acclimation capacity and thermal variability in marine, freshwater and terrestrial habitats. Our results are consistent with Gunderson and Stillman (2015) who show no relationship between plasticity in heat tolerance and latitude or thermal seasonality. However, other analyses on heat tolerance limits have found relationships between latitude (a proxy for seasonality) (Morley et al., 2019) or even direct measures of thermal variability (Verberk et al., 2024). Seebacher et al. (2015) also found that acclimation capacity was related to a population's thermal variability; however, relationships depended on the habitat and traits in question, and tropical animals showed greater acclimation capacity. Discrepancies across studies could be related to the taxa included in analyses (e.g. Morley et al., 2019), different traits or possibly the fact that different climate projections/models are being used to quantify thermal variability. Latitude covaries with a diversity of different ecological attributes aside from temperature (Louthan et al., 2021), which means latitude may be capturing other aspects of the environment that affect acclimation capacity. In addition, modelling realistic microenvironments across such diverse taxa is also challenging because it is unclear what the most appropriate spatial and temporal scale might be that is of evolutionary relevance. Historical temperature time series' may not be representative of the selective environment a population has experienced, making relationships between capacity for acclimation and temperature variability (or predictability) difficult to pin down.

5 | CONCLUSIONS

Enhanced knowledge of how variation in physiological rates varies across populations and species, and the degree to which they can be adjusted in response to the environment leads to more informed predictions about the ecological and evolutionary dynamics of natural populations (Cooke et al., 2021; Forsman, 2015; Sanderson et al., 2023; Seebacher et al., 2023). We show general patterns across taxa and habitats that provide a foundation to understand the relationship between plasticity and trait variance, as well as particular trade-offs that could impact the benefits (or lack thereof) of acclimation. It is important to recognise, however, that these patterns do not necessarily apply to all populations. Substantial variation in acclimation responses and changes in variance exist among populations and traits, as evidenced by wide prediction intervals and substantial study- and trait-level variance estimates, which is consistent with our understanding of factors influencing variation in performance curves across taxa (Rezende & Bozinovic, 2019; Tattersall et al., 2012). Conservation efforts are often targeted at particular populations or species, and taxonomic differences are important in this context. Regardless, quantitative measures of the changes in variance in physiological rates could be better incorporated into physiological and ecological models to provide more nuanced, and possibly more realistic, predictions about the impacts of climate change on natural populations. While we do not yet understand the relative contribution of environmental and genetic factors to variance changes, models could better decouple how different levels of heritability and total variance impact evolutionary and ecological predictions. Our meta-analysis now provides the opportunity to parameterise such models, and ensure they are better aligned with empirical findings.

Many fascinating questions remain unanswered that will require greater focus on the consequences of changes in variance (rather than just the mean). Particularly interesting questions include: How do differences in physiological rate variance change energy flow across trophic levels within communities? What are the biochemical, cellular and physiological mechanisms that underlie differences in physiological rate variance across habitats? Are changes in variance in one trait associated with changes in other traits, or do some traits increase while others decrease? Are changes in physiological rate variance correlated with changes in genetic variation? Answers to these questions will require integrative approaches that combine empirical and theoretical work across multiple levels of biological organisation but will likely provide useful advances in understanding the full consequences that climate change will have on ectotherms across major ecosystems globally.

AUTHOR CONTRIBUTIONS

Conceptualization: Daniel W. A. Noble, Fonti Kar, Frank Seebacher and Shinichi Nakagawa. Methodology: Daniel W. A. Noble, Alex Bush, Fonti Kar, Frank Seebacher and Shinichi Nakagawa. Investigation: Daniel W. A. Noble, Fonti Kar, Frank Seebacher and Shinichi Nakagawa. Visualisation: Daniel W. A. Noble. Supervision: Daniel W. A. Noble, Shinichi Nakagawa and Frank Seebacher. Writing—original draft: Daniel W. A. Noble. Writing—review & editing: Daniel W. A. Noble, Alex Bush, Fonti Kar, Frank Seebacher and Shinichi Nakagawa.

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CONFLICT OF INTEREST STATEMENT

The authors declare that they have no competing interests.

DATA AVAILABILITY STATEMENT

All data and code used to reproduce analyses can be found on GitHub at: https://github.com/daniel1noble/Q10_meta_analysis and is deposited in Zenodo (https://doi.org/10.5281/zenodo.11123600; Noble et al., 2024).

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SUPPORTING INFORMATION

Additional supporting information can be found online in the Supporting Information section at the end of this article.

Figure S1. PRISMA flow diagram of the literature search and screening process.

Figure S2. Bubble plot of the relationship between $\ln RR_{Q_{10}}$ and maximum temperature used in treatments within a study.

Figure S3. Bubble plot of the relationship between $\ln RR_{Q_{10}}$ and acclimation time for terrestrial (green), marine (orange) and freshwater (blue) habitats.

Figure S4. Mean acute $\ln RR_{Q_{10}}$ for cool (blue) and warm (red) acclimated populations for terrestrial (diamonds), marine (square) and freshwater (circle) habitats.

Figure S5. Meta-analysis results for $\Delta \ln RR_{Q_{10}}$ across different habitats.

Figure S6. Past climate variability did not predict acclimation responses as measured by $\Delta \ln RR_{Q_{10}}$.

Figure S7. Mean-standard deviation relationships for the acute and acclimation responses across all habitats.

Figure S8. Estimated mean acute and acclimation $\ln CVR_{Q_{10}}$ for marine, freshwater and terrestrial habitats.

Figure S9. Estimated mean acclimation and acute $\ln CVR_{Q_{10}}$ for tissue/whole-orgamism traits and biochemical traits across terrestrial (A), marine (B) and freshwater (C) habitats.

Figure S10. Estimated mean acclimation and acute $\ln CVR_{Q_{10}}$ for adult (a) and juvenile (j) life-history stages for terrestrial (A), marine (B) and freshwater (C) ectotherms.

Figure S11. Meta-analysis results for organismal and biochemical trait categories.

Figure S12. Acute and Acclimation $\ln RR_{Q_{10}}$ across detailed trait categories for (A) marine, (B) freshwater and (C) terrestrial systems.

Figure S13. Acute and acclimation $\ln VR_{Q_{10}}$ across traits for (A) marine, (B) freshwater and (C) terrestrial systems.

Figure S14. Acute and acclimation $\ln CVR_{Q_{10}}$ across traits for (A) marine, (B) freshwater and (C) terrestrial systems.

Figure S15. I^2 estimates.

Figure S16. Predicted mean acclimation (thick black line) $\ln RR_{Q_{10_{acclim}}}$ (A) and $\ln CVR_{Q_{10_{acclim}}}$ (B) as a function of the thermal predictability for wild populations across marine, freshwater and terrestrial habitats.

Figure S17. Funnel plot of precision (1/sampling standard error) against effect size for (A) log response ratio Q_{10} ($\ln RR_{Q_{10}}$), (B) log coefficient of variance ratio Q_{10} ($\ln CVR_{Q_{10}}$) and (C) log variance ratio Q_{10} ($\ln VR_{Q_{10}}$).

Figure S18. Simulated performance curves for $n=40$ individuals in four hypothetical scenarios with varying performance breadth (σ), optima (δ) and skewness (α).

Table S1. Slopes and 95% credible intervals (lower = 2.5% and upper = 97.5%) of log transformed standard deviation (log(SD)) and log transformed mean (log(mean)) for each of the four treatment types (r1.1, r1.2, r2.1, r2.2).

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